

Sec. 3.2 Set of sets

We have seen that a set can be an element of another set. For example, $\{x, y, \{1\}\}$

Definition. The **power set** of a set A , denoted by $P(A)$, is the set of all subset(s) of A .
Using set builder notation:

Example 1.

- If $A = \emptyset$, then $P(A) =$
- If $A = \{a\}$, then $P(A) =$
- If $A = \{a, b\}$, then $P(A) =$
- If $A = \{a, b, c\}$, then $P(A) =$

Theorem 2. If A is a finite set with n elements, then $P(A)$ has 2^n elements.

Reason:

Exercise 3. Let $X = \{0, \{1\}, \{x, y\}\}$.

- $|X| =$
- $P(X) =$
- $\{A \in P(X) : |A| = 2\} =$
- $\{A \in P(X) : 0 \in A\} =$

Exercise 4. Express each set using roster notation. Then give the cardinality of the set.

1. $P(\emptyset)$

2. $P(P(\emptyset))$

3. $P(P(P(\emptyset)))$

Exercise 5. Determine the truth value of the following statements. For each item from 13 to 16, if it is false, find a counterexample.

1. $\{a\} \subseteq \{x, y, a\}$

2. $\{a\} \in \{x, y, a\}$

3. $\{a\} \subseteq \{x, y, \{a\}\}$

4. $\emptyset \subseteq \emptyset$

5. $\emptyset \subseteq \{\emptyset\}$

6. $\{\emptyset\} \subseteq \{\{\emptyset\}\}$

7. $\{\emptyset\} \in \{\emptyset\}$

8. $\emptyset \subseteq \{\{\emptyset\}\}$

9. $\emptyset \in \{\{\emptyset\}\}$

10. $\emptyset \subseteq \{x, y, \emptyset\}$

11. $\emptyset \in \{x, y, \emptyset\}$

12. $\{\emptyset\} \subseteq \{x, y, \emptyset\}$

13. $(A \subseteq B) \wedge (B \subseteq C) \rightarrow A \subseteq C$

14. $(A \subseteq B) \wedge (B \in C) \rightarrow A \in C$

15. $A \subseteq B \leftrightarrow P(A) \subseteq P(B)$

16. $A = \emptyset \leftrightarrow P(A) = \emptyset$